

Drug Summary

Entrectinib (Rozlytrek) is a small-molecule tyrosine kinase inhibitor initially approved for adults with ROS1-rearranged non-small cell lung cancer and for adult and pediatric patients with solid tumors harboring NTRK1/2/3 gene fusions ¹ ². In breast cancer, entrectinib's use is targeted to tumors driven by oncogenic NTRK fusions – a rare subset comprising <1% of all breast cancers ³. Notably, >90% of secretory breast carcinomas (a rare histologic subtype) harbor an ETV6-NTRK3 fusion, making them exquisitely sensitive to TRK inhibition ³ ⁴. Clinical trial data confirm deep and durable responses in NTRK fusion-positive breast cancers: in the STARTRK-2 basket trial, entrectinib achieved an objective response rate ~70–83%, including multiple complete remissions, in heavily pretreated patients with metastatic NTRK-driven breast tumors ⁵ ⁶. These responses were observed across both secretory and non-secretory breast cancer cases and were notably sustained (median duration of response ~13 months) ⁷ ⁶. Entrectinib penetrates the central nervous system (CNS), an important feature for patients with breast cancer brain metastases – no CNS progressions were seen among trial patients with baseline brain metastases ⁸ ⁹. Overall, while not a standard therapy in most breast cancer subtypes, entrectinib is a precision-oncology option in the uncommon scenario of an NTRK fusion, offering substantial tumor regressions in that biomarker-defined population ¹⁰.

Identifiers & Synonyms

- **ChEMBL ID:** ChEMBL1983268 ¹¹
- **DrugBank ID:** DB11986 ¹²
- **Synonyms/Trade Names:** Rozlytrek (marketed brand name) ¹³; RXDX-101 (development code) ¹³; NMS-E628 (development code) ¹³.
- **Other identifiers:** CAS Number 1108743-60-7; PubChem CID 25141092. (Entrectinib is sometimes referred to as a “pan-TRK inhibitor with ROS1/ALK activity” in literature.)

Mechanism of Action (Breast Cancer Context)

Entrectinib is an ATP-competitive inhibitor of the tropomyosin receptor kinases TRKA, TRKB, and TRKC (encoded by NTRK1/2/3), as well as the ROS1 and ALK receptor tyrosine kinases ¹⁴. In NTRK fusion-positive breast cancers, these fusions (e.g. ETV6-NTRK3 in secretory carcinoma) produce constitutively active TRK chimeric proteins that drive oncogenic signaling ¹⁵. The unchecked TRK signaling activates downstream pathways such as the RAS-MAPK cascade, PI3K-AKT-mTOR, and PLCγ-PKC, leading to increased cell proliferation and survival with suppressed apoptosis ¹⁶ ¹⁵. Entrectinib's inhibition of TRK kinases in this context blocks these downstream pro-growth signals, thereby inducing cell-cycle arrest and apoptosis in the tumor. Clinically, this translates into tumor shrinkage in TRK-driven breast cancers: for example, secretory breast carcinoma cells, which depend on NTRK3 fusion signaling, undergo rapid tumor cell death when TRK is inhibited ⁴. Importantly, entrectinib's CNS penetrance allows it to inhibit TRK/ROS1/ALK signaling in brain lesions as well ¹⁷ ³. While NTRK fusions are agnostic to hormone receptor status, in practice most occur in triple-negative tumors (secretory type); thus, TRK inhibition often fits into a TNBC treatment context ⁴ ¹⁸. Entrectinib's multi-kinase activity against ALK and ROS1 is less frequently relevant in breast

cancer (ALK/ROS1 fusions are extremely rare in breast malignancies), but illustrates its broader mechanism: by targeting key oncogenic drivers, entrectinib can override diverse growth signals. In summary, in an NTRK-driven breast cancer, entrectinib disables the primary driver pathway (TRK signaling) to halt tumor growth and promote apoptosis ¹⁹, offering a targeted therapy approach analogous to HER2 or ER targeting in other breast cancer subsets.

Primary Targets (Human)

Entrectinib's direct molecular targets in humans are the kinase domains of:

- **NTRK1 (TRKA):** Neurotrophin receptor kinase A. Entrectinib binds the TRKA kinase domain to prevent neurotrophin (NGF)-mediated signaling ¹⁴.
- **NTRK2 (TRKB):** Neurotrophin receptor kinase B. Inhibition of TRKB disrupts BDNF/NT-3 growth factor signaling, curtailing downstream survival pathways ¹⁴.
- **NTRK3 (TRKC):** Neurotrophin receptor kinase C. Entrectinib blocks the TRKC kinase, which in fusion form (most commonly ETV6–NTRK3) is an oncogenic driver in secretory breast carcinoma ¹⁴ ⁴.
- **ROS1:** Proto-oncogene tyrosine-protein kinase ROS1. Entrectinib inhibits ROS1, a receptor tyrosine kinase; while ROS1 fusions are common in lung cancer, they are very rarely found in breast cancer ¹⁴. (Its inclusion as a target is relevant to the drug's lung cancer indication.)
- **ALK:** Anaplastic lymphoma kinase. Entrectinib also targets ALK, another RTK frequently rearranged in other cancers (e.g. ALK fusions in lung cancer) ¹⁴. ALK fusions are exceedingly uncommon in breast cancer, so this target is generally not engaged in the breast context.

Each of the above targets has been validated with biochemical binding assays and cellular studies ¹⁴. By inhibiting TRKA/B/C, ROS1, and ALK, entrectinib effectively shuts down multiple oncogenic signaling routes. (Notably, entrectinib has minor off-target activity on JAK2 and ACK1 kinases in vitro ²⁰, but the primary clinically relevant targets in breast cancer are the TRK family kinases.)

Pathways (Overview)

Entrectinib perturbs several key cellular pathways in breast cancer, primarily those downstream of TRK signaling, as well as adaptive pathways that modulate drug sensitivity or resistance. The main pathways affected include:

- **TRK Oncogenic Signaling:** *Neurotrophin/TRK receptor pathway* – This is the core driver pathway in NTRK-fusion breast cancers. Ligand-independent TRK activation (via fusion) leads to robust downstream signaling through RAS/MAPK, PI3K/AKT, and PLCγ pathways ¹⁶. Entrectinib's mechanism-of-action is to downregulate this pathway by inhibiting TRK kinases, which is mechanistically linked to its anti-tumor effect (pathway suppression is required for tumor sensitivity) ¹⁵ ¹⁹. This pathway is *pan-breast cancer* relevant only in the rare fusion-positive cases, not in most breast tumors.
- **RAS–MAPK/ERK Pathway:** A key proliferative signaling cascade often activated by TRK. Entrectinib indirectly suppresses the MAPK/ERK pathway by blocking upstream TRK/ALK signals ¹⁶. Downregulation of MAPK signaling is associated with tumor response (sensitivity), whereas re-

activation of RAS/MAPK (via secondary mutations or bypass tracks) can drive resistance ²¹. This pathway's modulation can be both *mechanistically linked* (part of TRK's immediate downstream) and *resistance-associated* (when upregulated as a bypass mechanism).

- **PI3K-AKT-mTOR Pathway:** Another major survival pathway under TRK control. TRK fusions constitutively activate PI3K/AKT signaling, promoting survival and growth ¹⁶. Entrectinib-mediated TRK inhibition attenuates the PI3K-AKT pathway; effective downregulation correlates with drug sensitivity. However, if the PI3K pathway is independently activated (e.g. by a PIK3CA mutation or PTEN loss), it can blunt the drug's efficacy (i.e. an *upregulated PI3K pathway can confer resistance*) ²². This pathway is *mechanistically linked* (directly downstream of TRK) and also emerges as a *resistance-associated* pathway if aberrantly activated outside of TRK control.
- **PLCγ-PKC Pathway:** TRK activation stimulates phospholipase C-γ, leading to PKC activation and calcium signaling ¹⁶. This pathway influences cell motility and survival. Entrectinib likely reduces PLCγ signaling by TRK inhibition. While less commonly discussed in breast cancer, it may play a role in the biology of NTRK-driven tumors. (Pathway identifiers: e.g. *REACTOME: PLC-gamma1 signaling*). Suppression of this pathway is part of the immediate drug effect (*mechanistically linked*), though its role in resistance is not well characterized in breast cancer.
- **Cell-Cycle Progression (E2F/G2-M):** By inhibiting key growth signals, entrectinib causes cell-cycle arrest. NTRK fusion-driven cells typically exhibit high proliferation rates ¹⁵; treatment leads to downregulation of cell-cycle gene programs (e.g. E2F targets, mitotic regulators) and accumulation of cells in G₁. This *downregulated proliferative pathway* is associated with drug *sensitivity*, as halting cell-cycle progression is required for tumor regression. Conversely, failure to fully suppress cell-cycle drivers could allow some cells to continue dividing (a potential resistance factor).
- **Apoptosis Pathway:** TRK-driven tumors actively suppress apoptosis ¹⁵. Entrectinib shifts the balance toward pro-apoptotic signaling, in part via downregulating MAPK/PI3K survival signals ²³. Upregulation of apoptosis (e.g. increased BIM, caspase activation) is a hallmark of sensitive response (drug-treated cells undergo apoptosis) ¹⁹. If apoptotic pathways are impaired or anti-apoptotic proteins (BCL-2 family) remain high (*apoptosis downregulated*), the tumor may resist cell death despite TRK inhibition (thus *resistance-associated* if apoptosis is not engaged).
- **MET Signaling Pathway:** The hepatocyte growth factor (HGF)/MET receptor pathway is not directly targeted by entrectinib, but it has emerged as a critical *bypass resistance pathway*. Off-target resistance to TRK inhibitors often involves upregulation of MET signaling (e.g. MET gene amplification) which can drive tumor growth independent of TRK ²⁴. Thus, *upregulated MET pathway* is *resistance-associated*. In the absence of such upregulation, MET is not prominent in NTRK-driven breast tumors; however, if MET becomes activated, it provides an alternative survival route that diminishes drug efficacy.
- **JAK/STAT Signaling:** ALK fusion signaling (in non-breast contexts) activates JAK/STAT, and TRK signaling can intersect with cytokine pathways. Entrectinib's inhibition of ALK (if ever relevant in breast cancer) would downregulate JAK/STAT signals ¹⁶. While not a primary pathway in NTRK-fusion breast cancer, increased JAK/STAT signaling (through inflammatory cytokines or other RTKs) could hypothetically support tumor survival if TRK is blocked. This would be considered *resistance-associated* if observed (e.g. IL6/STAT3 upregulation promoting cell survival). Clinically, this is more

relevant in other tumors; no direct breast cancer evidence yet, so it remains a potential adaptive pathway.

- **Immune Microenvironment/TME:** NTRK fusion-positive breast cancers often have low tumor mutational burden and are not heavily immunogenic ²⁵. Entrectinib's primary effect is tumor-intrinsic, but an indirect influence on the TME may occur as tumors shrink (e.g. increased immune cell infiltration in regressing tumors). There is no definitive human breast data on immune signatures changing with TRK inhibition. However, some NTRK-driven tumors express PD-L1 (about 25% in one series) ²⁶. The presence of an active immune pathway is not required for entrectinib efficacy, but could modulate long-term outcomes. If anything, immune pathways are considered *subtype-specific* (e.g. some triple-negative tumors) but not the main factor in TRK inhibitor response.

In summary, entrectinib in breast cancer primarily downregulates the *oncogenic TRK-RAS-MAPK-PI3K axis* (driving tumor shrinkage), and secondary pathways like *apoptosis* are upregulated as a consequence of effective treatment. Resistance tends to involve *upregulation of bypass pathways* (e.g. RAS/MAPK reactivation, MET signaling) or insufficient *downregulation of survival pathways* (e.g. persistent PI3K or anti-apoptotic signaling). The following sections detail these pathway regulations by context and subtype, with specific examples of sensitivity versus resistance mechanisms.

Upregulated Pathways (Drug Response or Resistance)

Observed Upregulations: In the context of entrectinib exposure, certain pathways can become upregulated as an adaptive response, especially in resistant tumors. Notably, re-activation of the RAS/MAPK pathway has been observed as a mechanism of acquired resistance. Preclinical models of NTRK1-driven cancer showed that tumors escaping TRK inhibition often had *increased MAPK/ERK signaling*, which drove proliferation despite continued drug treatment ²¹. Clinically, analyses of TRK inhibitor-resistant cancers (mostly outside breast cancer due to rarity) have found *upregulation of bypass RTK pathways* such as MET. MET gene amplification leads to heightened MET signaling, which can substitute for the blocked TRK pathway and was identified in multiple cases of off-target resistance to entrectinib ²⁴. Thus, an *upregulated MET pathway* is a hallmark of resistance in TRK-fusion tumors. There is also evidence that other growth factor receptors (EGFR, IGF1R, etc.) can be upregulated in tumor cells under the selective pressure of targeted kinase inhibition, providing alternative routes for activation of downstream pathways (though specific reports in NTRK+ breast cancer are lacking, this paradigm is seen in analogous settings ²⁷).

In responders, on the other hand, pathway upregulation is not prominent – the desired effect is pathway suppression. However, one potentially beneficial upregulation in sensitive contexts could be immune-related: as tumor cells die, there may be *upregulation of antigen presentation and interferon pathways*, recruiting immune cells to help clear the tumor. While speculative in breast cancer, this pattern (immune activation signature) has been noted in other cancers during effective targeted therapy responses. No breast-specific data has confirmed this for entrectinib, so the main known upregulations remain those linked to resistance.

Subtype context: Upregulated resistance pathways appear to be *pan-tumor mechanisms* rather than specific to a breast cancer subtype, given the low numbers of NTRK+ cases. For instance, MET amplification

has been noted as a resistance mechanism in various solid tumors treated with TRK inhibitors ²⁴ . If an NTRK-fusion breast cancer were to progress on entrectinib, one would investigate for similar *bypass pathway upregulation* (such as MET or MAPK pathway mutations) regardless of subtype. One context-specific note: in an HR+ breast cancer case with an NTRK fusion (rare, but reported after endocrine therapy failure ²⁸), there might be upregulation of estrogen receptor signaling as an alternate driver. If endocrine therapy is withdrawn, ER pathway could re-engage; however, if the tumor's growth is primarily fusion-driven, ER upregulation alone may not fully restore proliferation. Still, combination strategies (e.g. continuing endocrine therapy) could be considered to avoid upregulation of that pathway as a resistance route. Similarly, in HER2-positive breast cancer, upregulation of HER2 signaling is a concern if a co-existing NTRK fusion is being targeted – the HER2 pathway might dominate unless also inhibited. Indeed, a reported breast tumor harbored an ERBB2–NTRK3 fusion alongside HER2 amplification ²² , underscoring that multiple pathways can co-drive tumor growth. In such a case, HER2 pathway upregulation would need concurrent targeting (e.g. HER2-directed therapy) along with entrectinib to achieve response.

In summary, **pathway upregulation is mainly detrimental** in the context of entrectinib therapy, as it usually signifies compensatory survival signaling leading to drug resistance. Key examples are the **MAPK/ERK pathway** and **MET signaling**, whose upregulation in breast cancer cells can bypass TRK inhibition and drive continued tumor growth ²¹ ²⁴ . Monitoring for these changes (via genomic or transcriptomic profiling at progression) is critical, as it may guide combination therapies (e.g. adding a MET inhibitor or an ERK inhibitor to re-suppress the upregulated pathway).

Downregulated Pathways (Drug Response or Context)

Entrectinib's therapeutic effect in NTRK-driven breast cancer is chiefly through *downregulation of the oncogenic pathways* that the tumor depends on. The most directly impacted pathway is the **TRK signaling axis** itself: drug binding leads to an acute reduction in TRK kinase activity, which downstream manifests as suppression of the RAS–MAPK and PI3K–AKT cascades ²⁹ . This *transcriptional and signaling downregulation* is evident in treated tumors – for instance, genes in the cell proliferation cluster (Cyclins, E2F targets) get downregulated as cell cycling slows, and pro-survival mediators (like phosphorylated AKT/ERK levels) drop, tilting the cells toward apoptosis ¹⁵ ¹⁹ . In patients who respond to entrectinib, one would expect to see decreased expression of proliferation-associated gene sets and restoration of pro-apoptotic factors (which were previously kept low by TRK-driven signals). Indeed, *induction of apoptosis* can be viewed as a downstream consequence of multiple survival pathways being turned off; markers of apoptosis (cleaved caspases, etc.) rise as those pathways are downregulated ²³ .

Certain pathway downregulation, however, can be unfavorable. A prominent example is the loss of negative regulators of signaling. Under chronic TRK signaling, cells often upregulate feedback inhibitors (e.g. sprouty (SPRY) proteins or MAPK phosphatases like DUSP6) that restrain pathway activity to some degree. When entrectinib suddenly blocks TRK, those feedback elements might be downregulated (since their expression was driven by the pathway) and, if the drug does not fully eliminate the signal, the loss of feedback can paradoxically allow partial pathway reactivation. While specific data in NTRK+ breast cancer are lacking, this phenomenon is known in other RTK inhibitor contexts: for example, EGFR inhibition leads to loss of ERK feedback inhibitors, and any remaining upstream signal can then strongly reactivate ERK. By analogy, *downregulation of feedback regulators* could contribute to resistance in TRK-inhibited cells. This underscores

the importance of complete pathway suppression – incomplete TRK/MAPK downregulation may enable a rebound of signaling.

Another *harmful downregulation* is the loss of tumor suppressor function. If a breast tumor loses **PTEN** (a negative regulator of PI3K), the resultant unchecked PI3K activity can maintain AKT signaling even when TRK is inhibited. Functionally, PTEN loss = downregulation of the PI3K restraining pathway, which is equivalent to an upregulation of the PI3K pathway output. In NTRK+ cancers, co-mutation analysis found PTEN loss to be exceedingly rare ³⁰, but a similar concept is a PIK3CA gain-of-function mutation (effectively downregulating the dependency on upstream signals). One NTRK-fusion breast cancer case harbored a PIK3CA E545K mutation ²²; such a tumor might exhibit less pathway downregulation upon TRK inhibition because the PI3K pathway stays active from the mutation. Clinically, this could mean a less complete response. Thus, losing a restraining pathway (PTEN or other tumor suppressors) is *resistance-associated* as it prevents full downstream downregulation.

Subtype considerations: The pathways that need to be downregulated for response – namely proliferation and survival pathways – are universal across breast cancer subtypes. However, certain subtypes might have additional pathways in play. For HR+/HER2– tumors, hormone receptor signaling is a parallel pathway; while not directly affected by entrectinib, its continued activity could sustain some proliferation. Ideally, in an ER+ NTRK-fusion tumor, concurrent ER pathway downregulation (via endocrine therapy) would complement TRK pathway downregulation for maximal effect. If the ER pathway remains active (i.e. not downregulated), it could mitigate the impact of TRK inhibition. In HER2+ tumors, similarly, if HER2 signaling is not controlled, it could drive growth despite TRK being off. Therefore, in combined-driver scenarios, **lack of downregulation of the other oncogenic pathway** is a resistance risk. These nuances highlight that in *mixed-driver tumors*, multiple pathways must be downregulated to achieve a response (necessitating combination treatments).

In summary, **effective downregulation of TRK-driven signaling pathways is synonymous with sensitivity to entrectinib**, whereas failure or inability to downregulate key pathways (due to alternate drivers or loss of regulatory controls) corresponds with resistance. Entrectinib reliably induces *downregulation of the TRK–MAPK–PI3K signaling network* in NTRK-fusion breast cancer cells ²⁹; ensuring that no other pathway “stays on” is critical for durable response. If *some pathways are not fully turned off* – whether due to co-activations or adaptive feedback – the cancer may persist or recur. The next sections delve into specific mechanisms of sensitivity and resistance, framed by whether they involve up- or down-regulation of pathways.

Sensitivity Mechanisms

Several mechanisms and biomarkers of *sensitivity* to entrectinib have been identified, all converging on the principle that tumors most dependent on TRK signaling are the most sensitive. These include specific genetic events (like the presence of NTRK fusions without alternative drivers), pathway activity patterns (e.g. reliance on certain pathways that the drug effectively inhibits), and tumor biology contexts (such as lacking resistance alterations). Key sensitivity mechanisms in human breast cancer are:

- **NTRK Fusion as a Driver:** The foremost sensitivity determinant is the tumor having an *NTRK gene fusion that drives its growth*. Breast cancers harboring an NTRK fusion (especially ETV6–NTRK3) are highly responsive to TRK inhibition ³¹. This is evidenced by objective responses in essentially all evaluable NTRK-fusion breast cancer patients treated with entrectinib ⁵. The fusion is a gain-of-

function (GOF) event creating a dependency: the tumor cells become “addicted” to the TRK pathway. Consequently, they exhibit synthetic lethality to TRK blockade – removing this key signal causes cell death. **Subtype note:** This sensitivity spans any subtype if the fusion is the main driver. Secretory carcinoma (typically triple-negative) is the prime example, with dramatic responses observed ⁷. Even rare non-secretory cases (which might be HR+ or HER2+) responded if they had the fusion, indicating the fusion’s dominance over other signaling ³².

- **Lack of Redundant Oncogenic Drivers:** Tumors that do *not* harbor other strong oncogenic drivers (e.g. no concurrent HER2 amplification, no RAS/MAPK mutations, etc.) tend to be more sensitive. In practice, NTRK fusions are usually mutually exclusive with other major drivers ²⁷. For instance, in a genomic series, NTRK fusion-positive breast cancers had no co-occurring high-actionable alterations except one case with PIK3CA mutation ²². The absence of alternate drivers means the cancer cell’s survival is singularly reliant on the NTRK pathway – a setup that makes entrectinib highly effective. If a tumor did have dual drivers, the sensitivity would be diluted. Therefore, a *clean oncogenic background* (NTRK fusion as the sole driver) is a mechanism underlying deep sensitivity.
- **High TRK Pathway Output (Addiction):** Within NTRK-fused tumors, those that are truly “addicted” to TRK signaling – evidenced by high expression of TRK and strong activation of downstream pathways – are especially sensitive. For example, secretory breast carcinoma universally expresses the NTRK3 fusion protein at high levels (diffuse TRKC staining by IHC) ³³. This correlates with extreme dependency: the case report of a secretory carcinoma showed tumor regression with TRK inhibitor after standard chemo failed ³⁴. In contrast, if an NTRK fusion were expressed at low levels or in only a subset of cells, sensitivity might be less uniform. Thus, a *robust activation of the TRK downstream gene signature* (e.g. high pERK, pAKT that collapse when TRK is inhibited) is a marker of sensitivity. It implies that upregulating the TRK pathway was driving growth, and conversely its acute downregulation by drug triggers tumor collapse.
- **Synthetic Lethal Interactions / Co-dependencies:** Though not yet well-explored for TRK in breast cancer, we can extrapolate. If a tumor has a co-dependency that complements TRK inhibition, it could enhance sensitivity. One theoretical example: if a tumor has DNA repair deficiencies (like BRCA mutation), it might already be stressed; adding TRK inhibition could push it over the edge (though BRCA mutation primarily confers PARP inhibitor sensitivity, not clearly TRK). Another angle is immune co-dependency: an “immune-hot” tumor (with T cells present) might experience immune-mediated tumor cell killing once entrectinib induces cancer cell apoptosis (release of neoantigens). There is no direct clinical evidence for this in NTRK+ breast cancer yet, but it’s noted that in some patients, rapid tumor regression on TRK inhibitors can invoke inflammatory responses (tumor flare as immune cells clean up debris). Overall, synthetic lethal interactions specific to TRK inhibitors are still being researched; the strongest known factor remains the TRK fusion itself.
- **Subtype-Enriched Patterns:** If we consider breast subtypes: *TNBC* (basal-like) generally lacks targeted therapies, but a *TNBC* harboring an NTRK fusion is an exception – it becomes exquisitely sensitive to entrectinib. Indeed, one case report described a refractory *TNBC* patient who responded to TRK inhibition after the tumor was found to have an NTRK fusion ³⁵ ³⁶. Meanwhile, *HR+/HER2-* tumors rarely have NTRK fusions; however, one analysis noted NTRK fusions in some ER+ tumors post-endocrine therapy ²⁸. If present, those cells likely emerged because they provided a growth advantage once hormone signaling was suppressed, meaning they are highly TRK-dependent and would be sensitive to TRK blockade. So an HR+ tumor that shifts to an NTRK-driven state would

respond (though one must often continue endocrine therapy too for residual ER+ cells). For *HER2+* tumors, as mentioned, sensitivity to TRK inhibitor would require concurrently controlling HER2 if it's also active; otherwise the benefit may be partial. But at least one *HER2+/NTRK3*-fusion case was identified ²² – such a tumor might need both entrectinib and anti-HER2 therapy for maximum effect (sensitivity in that scenario is multi-factorial). In summary, NTRK fusion-positive tumors across subtypes share the key sensitivity driver (TRK), but co-targeting other subtype drivers may be necessary in combination subtypes.

Priority Sensitivity Pathways for Upregulation: These refer to pathways whose increased activity correlates with greater sensitivity to the drug (paradoxically, upregulation makes the tumor more killable by the drug). This concept mainly applies to the target pathway itself and apoptosis:

- *TRK signaling pathway (upstream of drug target):* It may seem counterintuitive, but a tumor with *highly upregulated TRK signaling* is actually *more sensitive* to entrectinib – because the tumor is so reliant on that pathway. In other words, the **degree of pathway addiction** is proportional to sensitivity. A striking breast cancer example is secretory carcinoma: the ETV6-NTRK3 fusion is the defining lesion and is active in driving signaling; these tumors, in effect, have “upregulated TRK signaling” compared to typical breast cancers, and they respond dramatically to TRK inhibition ³¹ ⁴. The biological rationale is that when the dominant oncogenic pathway is maximally ON, blocking it creates a large disruption that the cancer cannot compensate for, leading to cell death. Thus, **upregulation of the oncogenic driver pathway** (TRK/RAS/AKT cascade) signifies a single-point dependency, making the tumor highly sensitive (a classic oncogene-addiction paradigm) ¹⁵ ⁴.
- *Apoptotic readiness:* If a tumor has a primed apoptotic program (e.g. low expression of anti-apoptotic proteins like BCL2, or a high baseline of pro-apoptotic factors), it will die more readily once growth signals are removed. In such cases, one could say the **apoptosis pathway is “upregulated” or poised**, facilitating drug-induced cell death. While not often described in these terms in clinical studies, preclinical research indicates that cancer cells with certain pro-apoptotic biases (such as high BIM) undergo apoptosis more quickly upon kinase inhibitor treatment ²³. In a breast cancer context, if an NTRK-driven tumor doesn't have additional lesions in apoptosis regulators (e.g. TP53 is wild-type or no MCL1 amplification), it may be more primed to execute cell death when TRK signaling is cut off. Therefore, **upregulation of apoptosis signaling** (or simply a cell state favoring apoptosis) increases sensitivity to entrectinib. The rationale is straightforward: entrectinib's job is to induce cancer cell death by shutting off survival pathways, and if the apoptotic machinery is ready to engage, the outcome is more efficient tumor cell clearance. (Clinically, this might manifest as some tumors showing rapid shrinkage vs. others only slow shrinkage or stable disease – the former likely had more robust apoptotic response.)

Currently, beyond these, there are no specific external pathways that one would intentionally boost to increase sensitivity to entrectinib in breast cancer. Unlike some treatments where, say, upregulating immune response can improve efficacy, TRK inhibitors don't have known sensitizing co-pathways that require activation. The emphasis remains on the tumor's innate dependency on the target pathway. As a corollary, combination therapies to *enhance sensitivity* often aim to **downregulate resistance pathways** rather than upregulate something (see below).

Priority Sensitivity Pathways for Downregulation: These are pathways where *decreased* activity leads to greater sensitivity to entrectinib. In practice, they are the pathways that – when suppressed – leave the tumor more vulnerable to TRK blockade. Key examples:

- **PI3K/AKT pathway:** Downregulation of PI3K/AKT signaling enhances reliance on the TRK pathway, thereby increasing sensitivity. If a tumor has low PI3K activity (no PIK3CA mutations, intact PTEN), it means TRK-MAPK is the primary survival route. Upon TRK inhibition, there isn't an alternate AKT survival signal, so cells undergo apoptosis. Supporting evidence comes from clinical observations that **co-activation of PI3K (the opposite of downregulation) can cause drug resistance**, implying that when PI3K is *not* activated (i.e. effectively down), the drug works better ²². Furthermore, preclinical studies in ALK-driven cancers (a similar RTK scenario) show that combining an AKT or mTOR inhibitor with the RTK inhibitor can produce deeper responses – essentially by *forcing downregulation of the PI3K/AKT pathway*, you eliminate a potential escape route. Thus, we infer that a breast tumor with inherently low PI3K signaling (or one treated with a PI3K pathway inhibitor concurrently) would be more sensitive to entrectinib. Biologically, **removal of PI3K-mediated survival signals** makes the cell wholly dependent on TRK → its shutdown by drug triggers cell death with little resistance.
- **MAPK/MEK/ERK pathway:** Similarly, downregulation of the MAPK pathway increases sensitivity to TRK inhibition. There is direct evidence: combining a MEK1/2 inhibitor (cobimetinib) with entrectinib in NTRK-driven models *dramatically delayed resistance* and led to more sustained tumor regressions ³⁷ ³⁸. This works by *ensuring the MAPK pathway stays suppressed*. In single-agent entrectinib treatment, if any RAS-MAPK reactivation occurs, it can drive survival; but if MAPK signaling is concurrently kept down (via MEK inhibition), the cancer cells have no escape. Therefore, a tumor where MAPK pathway is inherently sluggish or is pharmacologically downregulated is more sensitive. In breast cancer patients, this concept might translate to using MEK inhibitors for cases that start to show MAPK reactivation, or recognizing that a tumor without mutations in the MAPK pathway is likely to respond better. It's essentially the flip side of the resistance coin – *lack of MAPK activity beyond TRK* makes TRK inhibition more lethal to the cell.
- **Alternate RTKs and Survival Pathways:** Downregulation (or absence) of other receptor tyrosine kinase signaling also boosts sensitivity. For example, if a tumor does not have active HER2 or EGFR signaling (which most NTRK-fusion tumors do not), it is solely reliant on TRK. If it did, blocking those (downregulating their pathways) would increase reliance on TRK and thus make TRK blockade more effective. A hypothetical scenario: an NTRK-fusion, HER2-positive breast cancer – treating with both entrectinib and an anti-HER2 (to downregulate HER2 pathway) could yield a better response than entrectinib alone, because the HER2 pathway no longer provides a partial escape. In essence, **the more pathways are downregulated (except the target one)**, the more pressure is put on the target pathway; when that is blocked, the tumor has nowhere to turn. This is why combination therapies often aim to shut off parallel signals. In triple-negative/NTRK-fusion tumors, this is naturally achieved as they usually don't have other dominant pathways. In HR+ or HER2+ contexts with an NTRK fusion, downregulating the hormone or HER2 pathway via standard therapies likely increases the tumor's dependency on NTRK, thereby increasing sensitivity to entrectinib.

To summarize, *sensitivity is maximized when the oncogenic TRK pathway is the sole driver and all other potential survival pathways are minimized*. Upregulation of the TRK driver itself and a primed apoptotic response characterize highly sensitive tumors, whereas downregulation of any alternate survival circuits (PI3K, MAPK,

etc.) removes fallback options, also heightening sensitivity. These principles are leveraged in clinical practice by patient selection (testing for NTRK fusions to find the addicted tumors ¹⁰) and potentially by combining therapies to ensure no parallel pathway rescues the cancer cell.

Resistance Mechanisms

Despite initial responses, tumors can develop resistance to entrectinib through various mechanisms. In breast cancers with NTRK fusions, resistance may be less commonly observed (given many patients can achieve durable remission and the rarity of cases), but lessons from aggregated data across tumor types and case reports point to both *on-target* and *off-target* resistance pathways. Key resistance mechanisms include:

- **On-Target Mutations in NTRK:** A classic mode of acquired resistance is the emergence of mutations in the NTRK gene that prevent drug binding. These are typically solvent-front or gatekeeper mutations in the TRK kinase domain (e.g. NTRK1 G595R or G667C mutations) that sterically hinder entrectinib ³⁹. Such mutations maintain TRK signaling in the presence of drug, rendering the tumor refractory. While most data come from other cancers, there is no reason breast cancer would be different – if a patient remains on therapy long enough, tumor clones with NTRK kinase mutations can be selected. Indeed, next-generation TRK inhibitors (selitrectinib, repotrectinib) have been developed to overcome these on-target mutations ⁴⁰. So, **point mutations in the TRK kinase domain** are a primary resistance mechanism (on-target, hard resistance). Notably, these mutations are generally *subtype-agnostic* – the solvent-front mutation confers resistance in any NTRK3-driven tumor, be it breast or otherwise.
- **Activation of Bypass Pathways:** The tumor can activate alternate signaling routes to bypass TRK inhibition. We discussed MET amplification as one example – by increasing MET signaling, the cancer can drive MAPK/PI3K downstream pathways without TRK. In the pooled analysis of TRK inhibitor resistance, **MET amplification was one of the most commonly observed off-target mechanisms** ²⁴. Similarly, mutations in KRAS or activation of the MAPK pathway have been seen: e.g. a resistant ROS1-rearranged lung cancer on entrectinib had a KRAS G12C mutation arise ²¹. For NTRK, a case of neuroendocrine carcinoma (not breast) had a KRAS mutation with NTRK fusion as co-resistance mechanism ⁴¹. In breast cancer, one might suspect a PI3K pathway activation (such as that PIK3CA E545K in one patient ²²) could serve as a bypass – the tumor can continue to signal through AKT/mTOR even if TRK→RAS is blocked. Broadly, any pathway that converges on cell survival/proliferation and is not inhibited by entrectinib can mediate resistance if upregulated. **Common bypass pathways:** MET, EGFR family, IGF1R, SRC, or downstream components like KRAS/MEK. In breast cancer, the likely culprits are MET or other RTKs (since EGFR/HER2 would probably have been known if present from the start, except in those combination subtype cases). Bypass activation is often *associated with pathway upregulation* detected by either gene amplification or high ligand levels. This is why resistant biopsies or liquid biopsies are recommended; for example, if a progressing NTRK-fusion breast tumor shows MET amplification on sequencing, one can attribute resistance to that and consider adding a MET inhibitor.
- **Phenotypic Changes (EMT):** Epithelial-mesenchymal transition (EMT) is a process where cancer cells become more mesenchymal, motile, and drug-tolerant. There is evidence in other cancers that activation of TRK signaling can induce EMT and confer resistance to other TKIs ⁴². Conversely,

prolonged inhibition might select for more stem-like, mesenchymal cells. For ALK inhibitors in lung cancer, EMT has been observed as a resistance mechanism in a subset of cases (though not the dominant one) ⁴³. In breast cancer, especially basal-like, an EMT shift could reduce dependence on the original driver and increase intrinsic resistance (mesenchymal cells often rely on alternative pathways like AXL, IGF1R, etc.). While not reported yet for TRK inhibitor resistance in breast cancer (due to limited cases), EMT should be considered. If a resistant tumor biopsy shows loss of epithelial markers (E-cadherin) and gain of mesenchymal markers (vimentin), that suggests an EMT-driven drug resistance phenotype. This mechanism is more *general and context-dependent* – it could occur in triple-negative backgrounds more readily, since they have more plasticity. EMT itself doesn't pinpoint a single pathway but implies broad transcriptional reprogramming that can attenuate the apoptotic response to the drug and invoke alternate tyrosine kinases.

- **Drug Efflux and Pharmacologic Resistance:** Although entrectinib is not a strong P-gp substrate ⁴⁴, tumor cells could theoretically upregulate drug efflux pumps (like ABCB1 or ABCG2) to reduce intracellular drug concentration. Acquired overexpression of P-glycoprotein has caused resistance to many TKIs. If a breast cancer cell population started pumping out entrectinib faster, the effective drug level at the target could drop, leading to resistance. There's no documented case of this in the clinic for entrectinib, but it remains a known mechanism of resistance to chemotherapy and some targeted drugs. Given entrectinib's design to avoid P-gp (explaining its CNS activity ⁴⁵), this might be a less frequent route, or it might manifest with the M5 metabolite (since M5 is a P-gp substrate ⁴⁴). Still, it's wise to consider if a resistant tumor shows high expression of MDR1 gene (ABCB1) – especially in contexts like the brain (blood-brain barrier P-gp could limit CNS efficacy over time).
- **Tumor Heterogeneity and Pathway Compensation:** Breast cancers can be heterogeneous. It's possible that a minor subclone without the NTRK fusion (or with an additional driver) existed and, under therapy, that clone dominates. For instance, if an NTRK-fusion tumor had a subpopulation of cells that were primarily PI3K-driven (say a subclone with PTEN loss), entrectinib would wipe out the TRK-dependent cells but spare the PTEN-null cells, which could then progress. This is more a *selection mechanism* than a mutation acquired – the net effect is resistance through clonal evolution. To combat this, combination treatments or upfront multi-target strategies are sometimes used. In an NTRK+ breast tumor, baseline sequencing to look for other alterations (like PIK3CA mutation) is important; if present, one might anticipate that clone causing early relapse and consider adding a PI3Kα inhibitor to the regimen.
- **Subtype-Specific Resistance Patterns:** There is limited data due to the rarity, but we can extrapolate: In *HR+* NTRK-fusion breast cancers, resistance might involve reactivation of ER signaling (the tumor finding a way to growth signal via ER even if TRK is blocked). For example, if endocrine therapy is stopped during TRK inhibitor therapy, any residual ER+ cells could proliferate. Thus continued hormone suppression is advised. In *HER2+* cases, not treating the HER2 concurrently could allow HER2-driven resistance. Essentially, in multi-driver tumors, resistance can occur via the pathway relevant to that subtype. In *TNBC* (basal-like) NTRK-fusion cancers, aside from the above-mentioned MET or EMT mechanisms, another possibility is activation of immune evasion pathways – basal tumors often have some immune surveillance; effective targeted therapy might create immune pressure that selects for cells that have lost MHC expression or similar, allowing them to hide and regrow. This is speculative, but immune escape is a known resistance mechanism to therapies including targeted ones (less so than to immunotherapies, but still). If a resistant lesion has low immune infiltration whereas baseline did not, one could hypothesize that as a factor.

Now, focusing on the **prioritized resistance pathways by regulation**:

Resistance Pathways for Upregulation: These are pathways that, when upregulated, confer resistance to entrectinib. Based on current knowledge:

- *MAPK/ERK Pathway (up):* As mentioned, upregulation (reactivation) of RAS–MEK–ERK signaling is a major resistance route ²¹. This can happen via KRAS mutations, NRAS mutation, BRAF mutations, or loss of negative feedback. The practical outcome is that even with TRK blocked, the ERK pathway is firing, driving cell proliferation. In ROS1+ lung cancer treated with entrectinib, acquired KRAS G12C was seen and led to resistance by reactivating ERK ²¹. In NTRK+ contexts, cases of BRAF fusion or mutation have been noted post-larotrectinib (indicating similar pressure). Upregulated ERK signaling essentially bypasses the need for the upstream TRK/ROS1/ALK. Biologically, the tumor has found an alternate way to keep the growth engine on. In breast cancer, RAS pathway mutations are uncommon but not unheard of; if one emerges or is selected, it would upregulate ERK and cause resistance. This pathway is so critical that co-inhibition (TRK + MEK inhibitor) has been shown to forestall this resistance ³⁷. Thus, **MAPK pathway upregulation is a clear resistance mechanism** and is not subtype-limited (any cancer cell with unlocked RAS signaling will resist an upstream RTK inhibitor).
- *MET/HGF Pathway (up):* As discussed, MET amplification or ligand (HGF) overexpression leads to high MET activity which can drive both MAPK and PI3K pathways independent of TRK ²⁴. This is a documented cause of entrectinib resistance across tumor types; for example, a patient with NTRK fusion-positive colorectal cancer who progressed had MET amplification at resistance ²⁴. In breast cancer, MET is occasionally amplified especially in basal-like tumors or claudin-low subtype, though it's not common. If an NTRK+ breast cancer cell amplifies MET, the upregulated MET signaling will sustain tumor growth (through redundant downstream signaling) despite TRK inhibition. Targeting MET with a MET inhibitor could restore sensitivity in such a scenario – an approach analogous to overcoming MET-mediated resistance in EGFR-mutant lung cancers ⁴⁶. Upregulation of MET is thus a potent resistance pathway. One might test for MET changes if entrectinib stops working.
- *HER Family and Other RTKs (up):* Although not yet reported in an NTRK+ breast cancer case, it's plausible that upregulation of EGFR, HER2, or even FGFR could drive resistance. EGFR or HER2 activation would funnel signals through the same PI3K/ERK routes. For instance, if an initially NTRK-dependent cell upregulates EGFR (via EGFR mutation or increased EGFR ligand production), then EGFR could take over driving proliferation. Given that one NTRK3 fusion case in breast had concomitant HER2 amplification ²², if that tumor were treated with entrectinib alone, the HER2 pathway (upregulated by amplification) might continue tumor progression – effectively rendering the TRK inhibitor insufficient. Therefore, any *upregulated RTK signaling* axis can be a resistance mechanism. These tend to be context-specific: e.g., in an ER+ tumor, the RTK might be the ERBB family; in triple-negative, maybe EGFR or FGFR1. The principle remains that **increased activity of an alternate receptor pathway** gives resistance.
- *PI3K/AKT Pathway (up):* Upregulation of AKT signaling through either loss of PTEN or mutation of PIK3CA is a known general resistance factor to targeted therapies because it provides an independent survival signal. We consider this upregulation even though mechanistically it's often due to a *downregulation of PTEN* – the net effect is AKT is overactive. In a reported cohort, only 1 of 10 NTRK+ breast cancers had a PIK3CA mutation ²², but if resistance develops in that case, one

suspects the PI3K mutation blunted the drug's effect from the start. If a new PTEN loss or PI3K mutation arises, it would upregulate PI3K/AKT and thereby confer resistance. There's also the possibility of AKT hyperactivation by other means (feedback loops or enhanced IGFR signaling). In summary, **hyperactivation of the PI3K/AKT pathway** (via any mechanism) is an upregulation that can mediate resistance, as the cells no longer require the TRK input for survival signaling ¹⁶ . Clinically, combining entrectinib with an mTOR or AKT inhibitor could be a strategy if such upregulation is detected.

- *Cell Cycle Pathways (up)*: If the cell cycle checkpoints are lost, cells may proliferate through drug-induced stress. For instance, loss of RB or p53 can allow cells to continue cycling even when external signals are suboptimal. While not a "pathway" in the classical sense, the *upregulation of cell-cycle progression* (due to loss of tumor suppressors) can cause resistance by diminishing the drug's ability to induce arrest. A p53-mutant cell might not apoptose and can accumulate mutations, increasing the chance of other resistances. Many breast cancers (especially TNBC) have TP53 mutations inherently, which might explain why a minority of NTRK-fusion cases might not respond completely – if the cells can tolerate damage without dying. This is a broader context: *upregulated survival in the face of DNA damage or stress* (via loss of apoptotic checkpoint) overlaps with cell-cycle and apoptosis pathways. So one could list **p53/Cell-cycle checkpoint downregulation** (which leads to uncontrolled cell cycle = effectively an "up" of cell cycle in presence of drug) as contributing to resistance. It's less targetable but important mechanistically.

In essence, the *upregulated pathways that confer resistance* revolve around reactivating downstream signals or activating parallel drivers: MAPK, PI3K, MET, EGFR, etc. Each of these has been observed or has a strong rationale in TRK inhibitor resistance settings ²¹ ²⁴ . Monitoring and counteracting these (through combination treatments) is an area of ongoing research.

Resistance Pathways for Downregulation: These are perhaps less intuitive – pathways whose loss or suppression leads to a state of resistance. One way to think of this is *loss of dependencies that made the drug effective*. A prime example: loss of the NTRK fusion gene itself. Tumors can sometimes "lose" the fusion (e.g., by segmental deletion or by ceasing expression) under drug pressure, which sounds counterproductive because the driver is gone – but if the tumor cell isn't dead, it means it found a way to survive without that driver, possibly by shifting to another phenotype. In such a scenario, the **TRK pathway is effectively downregulated to zero**, and the tumor is no longer reliant on it – hence it's resistant to the inhibitor. This has been described as a mechanism in ALK or EGFR TKI resistance sometimes (small cell transformation or histologic change leading to loss of original target). There's no documented case of lineage switch in NTRK+ breast cancer, but one could imagine a tumor differentiating into a state where the fusion is not expressed (downregulated) and instead other pathways (like stem cell pathways) keep it alive. That would be a *downregulation of the target pathway itself resulting in resistance*.

More concretely:

- *Loss of feedback restraint (as touched on earlier)*: Downregulation of negative feedback loops (e.g. SPRY/DUSP) can lead to an overshoot of upstream signals once the drug pressure is reduced or intermittent. If drug levels dip (patient misses doses, etc.), the rebound could be strong and lead to progression. This isn't a typical acquired resistance mechanism, but a dynamic one – essentially the system is more unstable because feedback inhibitors are low. It underscores the importance of consistent dosing.

- *Immune surveillance downregulation:* If the immune system was helping control the tumor, and then the tumor finds a way to downregulate antigen presentation or interferon signaling, the immune system no longer keeps residual disease in check. For instance, loss of HLA expression would let some cells evade T cells. In the context of targeted therapy, this could allow minimal residual disease to persist and eventually grow. It's hypothetical for TRK inhibitors, but worth noting as a mechanism of resistance especially post-therapy when tumor cell population is diminished – selection might favor cells that can hide from immunity (which could be triggered by therapy-related cell death). So **downregulation of immune pathways** in tumor cells (or upregulation of PD-L1, which downregulates immune attack) might contribute to long-term resistance or recurrence. In breast cancer, immune evasion is particularly relevant in TNBC.
- *Downregulation of pro-apoptotic factors:* This is the flip side of the earlier sensitivity note. If a tumor acquires changes that *suppress apoptosis* – for example, upregulating BCL2 or MCL1 (which functionally downregulates apoptosis execution) – then even if TRK is inhibited, the cells may not die. Some resistant cancer cells avoid apoptosis by increasing anti-apoptotic protein levels; this is effectively **turning down the apoptotic pathway**. The result is drug resistance because the cells enter a drug-tolerant persister state rather than apoptosis. In chronic treatment, a subpopulation might do this. Therapeutically, adding a BCL2 inhibitor can sometimes kill those persisters.
- *Pathological downregulation due to tumor microenvironment:* Cancer-associated fibroblasts (CAFs) in the microenvironment can secrete factors that *downregulate the tumor's reliance on one pathway and induce broad drug tolerance*. For instance, HGF secretion by stroma can downregulate the need for EGFR in lung cancer (by activating MET). In breast cancer, certain stroma signals might reduce proliferation reliance on TRK and push cells into a slow-cycling mode. This overlaps with bypass, but mediated by environment – effectively the tumor cells behave as if the original pathway is less needed (quiescent or alternate signals from stroma). So one could say *downregulation of cell-autonomous signaling* (tumor cells becoming quiescent) is a resistance strategy – a drug-tolerant state.

In summary, “resistance pathways for downregulation” mostly refer to loss of factors that made the tumor sensitive. For entrectinib: loss of NTRK expression (target pathway off), loss of pro-apoptotic response, or loss of immune surveillance all fit this category. Each represents the tumor doing without something (target, cell death, immune help) to survive the drug. These mechanisms are often harder to tackle, as they're not just adding another signal that can be inhibited but rather fundamentally altering state (one might need to change treatment modality, e.g. switch to chemotherapy or immunotherapy if the tumor no longer expresses the target or is non-proliferative).

Breast cancer subtype-stratified resistance: Given the paucity of cases, we don't have subtype-specific resistance patterns documented for entrectinib in breast cancer, but we can speculate:

- In an *HR+ NTRK-fusion* tumor, if resistance emerges, check for estrogen receptor reactivation (ESR1 mutations, etc.) – because the HR+ subtype might resist by leaning back on ER signaling. That would be a subtype-specific angle (treat with endocrine therapy or CDK4/6 inhibitor).
- In a *HER2+ NTRK-fusion* tumor, resistance might simply be due to HER2 pathway dominance; ensure continued anti-HER2 therapy. If resistance still occurs, maybe the tumor downregulated NTRK (since

it had HER2 to rely on). One could see the NTRK fusion disappearing under pressure and the tumor behaving like ordinary HER2+ cancer again.

- In *TNBC (basal-like)*, one worries about aggressive biology: TP53 is usually mutated (apoptosis down), the microenvironment might be immunosuppressive, and there's greater plasticity. Resistance here may involve EMT or stem-like transitions, as mentioned. Also, TNBC could potentially activate alternate tyrosine kinases like FGFR2 or AXL (which have been seen in other contexts of TNBC resistance to targeted agents). Those wouldn't be subtype-exclusive but TNBC is more prone to such changes than luminal tumors.
- If the tumor had a *gBRCA mutation (HRD+)*, interestingly, one might consider whether prolonged targeted therapy selects for restoration of BRCA (as often happens with PARP inhibitor resistance). If BRCA function got restored, it wouldn't directly cause TRK inhibitor resistance, but it might make the tumor more genomically stable to accumulate other changes. That's tangential but worth noting in a broad sense of tumor evolution under therapy.

In conclusion, the resistance mechanisms to entrectinib in breast cancer mirror those observed in other solid tumors treated with targeted kinase inhibitors: the tumor either **hits back at the drug's target (mutating it)**, **finds a new growth pathway (upregulating an alternative)**, or **alters its phenotype to be less dependent or less vulnerable (through EMT, quiescence, etc.)** ³⁹ ²⁴. Each mechanism can potentially be addressed: next-gen TRK inhibitors for on-target mutations, combination therapies for bypass pathway activation, and possibly epigenetic or immunotherapy approaches for phenotypic changes. Ongoing trials and case studies will continue to illuminate how NTRK-fusion breast cancers escape control and how best to intercept those escape routes.

Breast Cancer Subtype–Stratified Evidence

HR+/HER2– (Luminal) Subtype: NTRK fusions are extremely rare in luminal breast cancers, but they have been identified. Notably, some ER+ breast cancers acquired NTRK fusions after progressing on hormonal therapy ²⁸. This suggests that under endocrine pressure, a subclone with an NTRK fusion may emerge and drive the tumor. In such cases, the tumor becomes less hormone-dependent and more TRK-dependent. Clinical evidence: a few ER+ patients in the pooled trials were included (the “non-secretory” NTRK-fusion cases). While detailed breakdown by ER status isn't published, responders did include non-secretory histology ⁴⁷, which could encompass luminal tumors. The implication is that even an ER+ cancer, if driven by an NTRK fusion, will respond to entrectinib, but optimal care might require combined endocrine therapy to suppress any residual ER-driven component. There is no indication that luminal tumors respond any worse to TRK inhibition – it's more about identifying the fusion. In practice, routine NTRK fusion testing in ER+ breast cancer isn't standard, but genomic profiling in metastatic cases could pick it up. If found, NCCN guidelines endorse TRK inhibitor therapy ¹⁰. Therefore, for luminal subtype, the evidence is that *presence of the fusion* is what matters; subtype per se does not prevent response, though one must manage the concurrent ER pathway.

HER2-Positive Subtype: Similarly uncommon for NTRK fusions. However, a striking case reported was a *HER2-enriched breast adenocarcinoma with an ERBB2–NTRK3 fusion and HER2 amplification* ²². This indicates the tumor had two drivers. There is no published report of treating such a case with entrectinib, but one can extrapolate: one would likely need to treat both drivers. Entrectinib alone might partially help (by blocking

the NTRK3 part of that fusion or any separate NTRK fusion transcripts), but without HER2 blockade the HER2 signaling could continue unchecked. If the fusion somehow involved the HER2 kinase domain fused to NTRK3, the biology is complex (it might have a chimeric function). General guidance: if a HER2+ patient is found to have an NTRK fusion, standard HER2-directed therapy remains necessary, and entrectinib could be added to target the fusion. The presence of the fusion might explain an unusual clinical behavior (for instance, maybe a HER2+ cancer not responding to usual treatment could have an NTRK fusion driving it). There's no direct trial evidence in HER2+ since such cases are anecdotal. But given that secretory carcinomas are typically HER2-, any HER2+ case is likely "non-secretory" category. In STARTRK-2, one patient had "breast NOS" which could have been HER2+; that patient's response was not evaluable ⁴⁷. It's hard to draw conclusions, but one might say: in HER2+ context, NTRK fusion is very rare, but if present, consider dual targeting.

Triple-Negative Breast Cancer (TNBC): This is the subtype where NTRK fusions, while still rare, are slightly more encountered aside from the distinct secretory carcinoma variant. A 2021 study specifically looked at TNBC for NTRK fusions using multiple assays ⁴⁸ ¹⁸. They found that true NTRK fusions were exceedingly rare (~0%–0.08% in non-secretory TNBC) ¹⁸. They did, however, identify one case that was initially in their TNBC cohort which turned out to be a misclassified secretory carcinoma (on morphology review) with an NTRK3-ETV6 fusion ³⁴. That case underscores that secretory carcinoma can be mistaken for conventional TNBC until genomic testing reveals the fusion. Secretory breast carcinoma is a special entity often considered a subset of TNBC because it lacks ER/PR/HER2, but it's clinically indolent and defined by NTRK3 fusion ⁴. In terms of evidence: secretory carcinomas (TNBC by receptors) have shown 100% ORR to TRK inhibitors in small series ⁴⁹. For *non-secretory TNBC* (basal-like, aggressive), only scattered case reports exist. One such case: a metastatic TNBC patient refractory to chemo was found to have an NTRK1 fusion (TPM3-NTRK1) and was treated with larotrectinib, achieving significant regression ⁵⁰. By analogy, entrectinib would also work. TNBC tumors often have co-alterations (TP53, etc.), but as long as nothing else is the dominant driver, they respond. TNBC may also have higher likelihood of something like MET amplification as a resistance (because basal-like tumors can amplify MET). So with TRK inhibitors in TNBC, vigilance for MET is warranted. In sum, TNBC patients, especially if of secretory histology or after multiple lines of therapy, should be tested for NTRK fusions as it opens a highly effective treatment option ⁵¹. The evidence base is limited but very promising for those few who have the fusion (deep responses in both secretory and a few non-secretory TNBC cases) ⁷.

gBRCA-mutated / HRD+ Subgroup: There isn't a known correlation between BRCA1/2 mutations and NTRK fusions – they appear to be independent events. Secretory carcinoma typically is not associated with BRCA mutations; it's more often sporadic. If a BRCA1-mutated patient's TNBC had an NTRK fusion, it would be coincidental (or possibly a therapy-related event after DNA-damaging treatment). Such a tumor might have dual susceptibilities: PARP inhibitors for the BRCA aspect and TRK inhibitor for the fusion. No reported case yet, but combination or sequencing of therapies could be considered (perhaps first use entrectinib to shrink tumor massively, then PARP inhibitor for remaining BRCA-mutant cells). HRD status by itself likely doesn't affect TRK inhibitor response, except insofar as HRD tumors might respond less to immunotherapy and more to targeted or chemo – in any case, if they have the fusion, the TRK inhibitor is the clear choice regardless of HRD.

Basal-like vs. Other Intrinsic Subtypes: Secretory carcinoma does not fit well into the typical PAM50 intrinsic subtypes (basal, HER2-enriched, luminal A/B) – it's its own category, often expressing basal cytokeratins but with a unique transcriptome. Basal-like breast cancers (PAM50 basal) usually correspond to conventional TNBCs. To date, essentially all NTRK fusion breast tumors have either been secretory (which

would probably cluster separately) or a smattering in others. One could guess that if a basal-like tumor did have an NTRK fusion, it might derive some basal-like gene expression from other background but would be heavily influenced by the fusion's transcriptomic footprint. The presence of an active TRK fusion might upregulate genes like ETV6–NTRK3 downstream targets, some of which could be related to neural or developmental pathways not typical in basal breast cancers. There isn't enough data to say how intrinsic subtype affects response – likely minimal effect, since the drug targets a pathway not directly tied to hormone/HER2 status.

Mammary Analog Secretory Carcinoma (MASC) of salivary gland vs. Secretory Breast Carcinoma: Just as a note, secretory carcinoma can occur in salivary glands (called MASC) with the same ETV6–NTRK3 fusion, and those respond well to TRK inhibitors too ⁵². It reinforces that histology with that fusion, regardless of site, is sensitive. So within breast, secretory carcinoma is almost a distinct disease embedded in the TNBC category that is exquisitely sensitive to TRK inhibition.

In conclusion, *subtype is not a barrier to efficacy* – the determinant is the presence of the targetable NTRK abnormality. However, subtypes inform what *else* might be going on: luminal tumors require concurrent endocrine management, HER2+ require HER2 therapy, TNBC might have more aggressive resistance patterns. Guidelines now reflect that any metastatic breast cancer, of any subtype, with an NTRK fusion should be considered for larotrectinib or entrectinib ¹⁰. This tumor-agnostic approach overrides subtype distinctions when it comes to TRK fusion-positive cases.

Contraindications and Safety

Entrectinib has **no absolute contraindications** listed by regulatory agencies aside from known severe hypersensitivity to the drug or its components ⁵³. In other words, there is no medical condition that universally forbids use; rather, clinicians must use caution and weigh risks in certain scenarios. Key safety and special population considerations include:

- **Cardiac Function (CHF):** Entrectinib can cause congestive heart failure in a small percentage of patients (3–4% experienced CHF in trials) ⁵⁴. The FDA label warns to assess left ventricular ejection fraction (LVEF) at baseline and monitor for signs of heart failure during treatment ⁵⁵. Patients with pre-existing serious cardiac disease (uncontrolled CHF, recent myocardial infarction, cardiomyopathy) were generally excluded in trials, so while not contraindicated, entrectinib should be used with caution in such patients. If a patient has severe heart failure at baseline, one might hold off or very closely monitor; the label doesn't outright contraindicate, but the risk is higher. Management includes periodic echocardiograms and prompt interruption if new CHF symptoms appear ⁵⁶. Thus, severe CHF could be considered a de facto contraindication until stabilized.
- **Central Nervous System Effects:** Entrectinib is CNS-active (which is good for metastases) but also CNS-penetrant in terms of side effects. It commonly can cause dizziness, cognitive changes, mood disorders, and sleep disturbances ⁵⁷. Patients are advised not to drive or operate machinery if these occur ⁵⁸. There's no specific contraindication for those with, say, baseline CNS disease or psychiatric illness, but one should be careful in patients with a history of debilitating CNS conditions (e.g. severe psychiatric disorders or baseline dizziness/balance issues) as entrectinib could exacerbate these ⁵⁷. Dose modifications might be needed if grade 3 neurotoxicity develops.

- **Skeletal Fractures:** An interesting safety finding is that entrectinib is associated with an increased incidence of skeletal fractures (observed in pediatric trials especially) ⁵⁹ ⁶⁰ . The mechanism isn't fully clear; it might relate to effects on bone growth pathways (since TRK fusions and their signaling can be involved in development). The label advises monitoring for fracture symptoms. This is particularly relevant for children and adolescents (whose bones are still growing) – in pediatric use, careful attention to bone health is needed. For adult breast cancer patients, this is less of a concern, though one might avoid concurrent use of medications that severely weaken bones (like long-term high-dose steroids) or at least monitor bone density if the patient is on therapy for a long time. No formal contraindication here, but a note: patients with osteoporosis or bone metastases at risk for fracture should be monitored.
- **Hepatic Impairment:** Entrectinib is primarily metabolized by the liver (CYP3A4 ~76%) ⁶¹ . The label indicates that moderate to severe hepatic impairment can increase drug exposure (~2-fold or more) ⁶² ⁶³ . While not contraindicated, in patients with *severe hepatic impairment* (e.g. Child-Pugh C cirrhosis), the pharmacokinetics are not well defined – caution is advised and dose reduction likely needed. The EMA SmPC likely recommends avoiding use in severe hepatic impairment if possible or only with great caution ⁶⁴ . No specific dose guidelines for Child-Pugh C are given (because it wasn't studied), so practically one might avoid entrectinib in a patient with end-stage liver disease due to unpredictable exposure. For *mild impairment*, no adjustment is indicated; for moderate, close monitoring is suggested ⁶⁵ ⁶⁶ . Liver enzyme elevations (AST/ALT) occurred in some patients (about 30–40%, mostly mild) ⁶⁷ , so periodic liver function tests are recommended.
- **Renal Impairment:** The drug is less affected by kidneys as it's mainly metabolized and fecally excreted. There's limited data in patients with creatinine clearance <30 mL/min. Since no formal contraindication is stated, entrectinib can be used in renal impairment with caution; the pharmacology suggests minimal unchanged renal excretion (so mild to moderate renal impairment likely doesn't significantly alter levels). In absence of clear guidance, severe renal failure patients should be closely monitored, but it's not a strict contraindication either.
- **Hyperuricemia/Tumor Lysis:** Entrectinib can cause hyperuricemia in 9% of patients, with rare cases of tumor lysis syndrome (TLS) reported ⁶⁸ ⁶⁹ . The drug causes rapid cell death in some patients (particularly those with high tumor burden responding dramatically), leading to uric acid and electrolyte disturbances. The label recommends checking uric acid before and during treatment and treating prophylactically with urate-lowering agents if levels rise ⁷⁰ ⁶⁸ . There was one TLS-related death in the clinical trials ⁷¹ , highlighting that risk. So in patients with massive disease burden (e.g. bulky tumors, high LDH), caution is needed at initiation – ensure hydration and consider allopurinol. Not a contraindication per se, but a strong precaution.
- **QT Interval Prolongation:** Entrectinib has been associated with QTc prolongation in some patients ⁷² . It's generally not severe (no high incidence of >500 ms reported publicly), but caution is warranted in those with congenital long QT or those on other QT-prolonging drugs ⁷³ ⁷⁴ . The label says to avoid concomitant use of drugs that prolong QT if possible ⁷⁵ . Monitor ECG periodically, especially in patients with risk factors (heart failure, electrolyte imbalances) ⁷⁶ . Again, not contraindicated, but if a baseline QT is very high, one might correct abnormalities first or choose alternative if arrhythmia risk is deemed too great.

- **Vision Disorders:** Blurred vision or other visual disturbances occurred (in about 14% of patients) on entrectinib, possibly due to off-target effects in the eye or CNS ⁷⁷. Patients should report vision changes; dose interruption or reduction can be done if severe. No contraindication, but patients with pre-existing severe eye conditions (like retinal degeneration) might be monitored by ophthalmologists during therapy.
- **Embryo-Fetal Toxicity: Use in pregnancy is contraindicated** due to the risk of fetal harm ⁷⁸ ⁷⁹. Entrectinib, like many TKIs, can cross the placenta and is teratogenic in animal studies (the label notes that based on TRK's role in development, disrupting it causes malformations) ⁷⁹. Women of child-bearing potential must use effective contraception during treatment and for at least 5 weeks after the last dose (per label) ⁷⁸. Men with female partners of child-bearing potential should use contraception for 3 months after treatment due to potential drug in sperm. Breastfeeding is also advised against during and 1 week after therapy, as the drug may be excreted in milk and harm a nursing infant (and also because of the systemic exposure). So while not literally labelled "contraindication," **pregnancy and lactation are strong no-go zones** for entrectinib ⁷⁸. If a patient becomes pregnant on the drug, it should be discontinued and the patient informed of potential fetal risk.
- **Pediatric Use:** Entrectinib is actually approved for pediatric patients ≥ 1 month old with NTRK-fusion solid tumors ⁸⁰, making it one of few TKIs with such a broad pediatric indication. In pediatrics, aside from the fracture risk and growth monitoring, main safety issues are similar. There is no contraindication by age; however, long-term effects on growth and development are an unknown, so continued surveillance is needed. For the context of breast cancer, pediatric use is not relevant (breast cancer in children is extremely rare; secretory carcinoma can occur in adolescents occasionally – in such cases, entrectinib could be used given approval).
- **Drug Interactions:** While not a classic "contraindication," co-administration of certain drugs is advised against. Notably, **strong CYP3A4 inducers** (like rifampin, phenytoin, St. John's Wort) can significantly reduce entrectinib plasma levels (by ~77% AUC reduction) ⁸¹, potentially leading to loss of efficacy – hence they should be avoided ⁸². Conversely, **strong CYP3A inhibitors** (like itraconazole, clarithromycin) can raise entrectinib levels ~6-fold ⁶⁴; if unavoidable, the entrectinib dose must be reduced (the label provides dose modification tables) ⁸³ ⁸⁴. Ideally, one would avoid strong inhibitors as well. **QT-prolonging drugs** (e.g. certain antiarrhythmics, fluoroquinolones) should be avoided concomitantly if possible, to reduce additive QT risk ⁷⁴. These interactions require careful management – not true contraindications, but co-administration is either *not recommended* or requires dose adjustment. Patients should also avoid grapefruit juice (a CYP3A inhibitor) during therapy ⁸⁵.
- **Special Populations:**
 - **Elderly:** No contraindication; efficacy and safety were similar in older adults, although slightly more sensitive to some toxicities might be possible.
 - **Performance status:** Very frail patients (PS 3-4) weren't in trials, but no absolute rule – just need risk/benefit consideration.
 - **Concurrent illness:** Patients with active uncontrolled infection, or recent surgery, etc., one would stabilize those conditions first (standard for any therapy).

- **Allergy:** If a patient had a severe allergic reaction to entrectinib (e.g. anaphylaxis) that would be a contraindication to re-challenge. There's no known cross-reactivity with other drugs except other TRK inhibitors (larotrectinib is different enough that allergy to one doesn't necessarily preclude the other, but caution).

Regulatory/guideline statements: - The FDA label explicitly has “**No contraindications**” listed ⁵³. - The EMA SmPC similarly does not list specific contraindications except standard ones (hypersensitivity to active or excipients). - NCCN and ESMO guidelines don't add contraindications beyond what labels state; they focus on patient selection (i.e. requiring the NTRK fusion) and note to manage toxicities. - ESMO recommendations for tissue-agnostic TRK inhibitors emphasize not to use these drugs in patients who lack the biomarker, and to be mindful of potential CNS effects and check heart function, but again, no unique contraindications. - One might consider that if a patient cannot swallow capsules (due to esophageal issues), entrectinib capsules can be opened and given as an oral solution or via feeding tube according to the label instructions ⁸⁶ – so route issues can be managed, not a contraindication.

In summary, **entrectinib is generally well tolerated**, and contraindications are mainly related to pregnancy and perhaps severe uncontrolled comorbidities. Key safety monitoring includes cardiac function (to avoid CHF), CNS symptoms, liver enzymes, uric acid, and ECGs for QT. These measures ensure that any emerging issues are caught early and managed (dose interruptions or reductions are effective in mitigating many side effects). For instance, if a patient develops grade 3 ALT elevation, one would hold drug until recovery and then resume at a lower dose. The drug's safety profile in the metastatic setting has been acceptable, with most adverse events being grade 1-2 (like fatigue, dysgeusia, dizziness) and manageable ^{68 77}. Both FDA and EMA have required post-marketing monitoring of some of these effects, but overall, *the benefits far outweigh risks in NTRK fusion-positive cancers*, provided that precautions are observed.

Trial and Guideline Context

Entrectinib's development in NTRK fusion cancers has been driven by *basket trials* and informed by tumor-agnostic drug approvals. The pivotal clinical evidence comes from an integrated analysis of three trials: ALKA-372-001 (an early-phase European trial), STARTRK-1 (phase I, global), and STARTRK-2 (phase II basket trial) ⁸⁷. In 2019, based on these studies, the FDA granted *accelerated approval* of entrectinib for **NTRK fusion-positive solid tumors** (regardless of site) and regular approval for ROS1+ NSCLC ⁸⁸. This made entrectinib one of the first tumor-agnostic therapies.

Clinical Trials in Breast Cancer: Breast cancer patients were included in STARTRK-2 (NCT02568267) as part of the basket. In that trial, seven breast cancer patients with NTRK fusions were treated ⁵. Updated results presented at ESMO Breast 2022 showed a 71% ORR in these patients (5 of 7 responded, with 2 CRs) ⁵. The median progression-free survival was around 10.1 months and median overall survival ~19–24 months in that cohort ^{89 6}. These outcomes are quite favorable given that these patients had metastatic disease, various prior therapies (some had ≥ 3 prior lines) ⁹⁰, and limited options. Notably, patients with secretory carcinoma (usually younger and with slow-growing tumors) achieved 100% ORR (all 4 secretory carcinoma patients responded, including the two CRs) ^{49 7}. Non-secretory cases had partial response or stable disease. No CNS progressions occurred among patients who had baseline brain metastases, highlighting entrectinib's CNS efficacy ⁹. Safety in this cohort mirrored the overall population – the most common adverse events were fatigue, dysgeusia, weight gain, dizziness; Grade ≥ 3 events were infrequent (one might note that weight gain and dizziness in this small breast subset were observed, consistent with the whole trial).

Another important trial was a *phase I pediatric study (STARTRK-NG)* which included a teenager with metastatic secretory breast carcinoma, who responded to entrectinib ⁹¹. This contributed to the pediatric approval.

Because of the rarity of NTRK fusions in breast cancer, there are no dedicated large breast-specific trials. Instead, evidence is pooled from these basket trials and case series. A noted case series (Drilon et al, New England Journal 2018 ⁴) included a patient with secretory breast carcinoma achieving a near-complete response to larotrectinib, underscoring the class effect. Entrectinib's efficacy is considered comparable to larotrectinib's; an indirect comparison in TRK-fusion cancers suggested similar ORRs and durability ⁹². However, entrectinib's unique point is its multi-target nature and CNS penetration.

Guideline Recommendations: International guidelines now incorporate NTRK fusion testing and TRK inhibitors: - The **NCCN** Guidelines for Breast Cancer (2023) state that for any unresectable/metastatic breast cancer that is NTRK fusion-positive (and has exhausted standard options), treatment with a TRK inhibitor (larotrectinib or entrectinib) is recommended ¹⁰. This appears as a footnote in the systemic therapy algorithms for all subtypes (often under the section for "Biomarker-driven therapy"). Specifically, the NCCN patients' guide lists: "For NTRK fusion, larotrectinib or entrectinib" as recommended treatments across the board ¹⁰. This affirms that, regardless of ER/PR/HER2, if an NTRK fusion is present, one of these targeted agents should be used. - The **ESMO** guidelines and ESMO Precision Medicine recommendations echo this: ESMO acknowledges NTRK fusions as an actionable alteration and advises testing especially in certain contexts (like secretory carcinoma or in tumors with wild-type for other drivers). In 2020 ESMO Translational guidelines, it was stated that TRK inhibitors have shown efficacy in NTRK-fusion breast cancers and can be considered after standard therapy or even earlier for secretory carcinoma given its unique biology ¹⁷. ESMO doesn't have a breast-cancer-specific line because of rarity, but in the ESMO "Recommendations for tissue-agnostic trials," TRK inhibitors are indicated for any NTRK-fusion solid tumor. - The **FDA** approval and **NCI** (NCCN) both emphasize that an FDA-approved companion diagnostic should be used to identify NTRK fusions (e.g. the FoundationOne CDx for solid tumors, or other NGS) ⁹³ ⁹⁴. In breast cancer, this means that comprehensive genomic profiling (which is increasingly done for metastatic cases) should include NTRK fusion analysis. Notably, IHC can be used as a screening for TRK protein, but as Wu et al showed, IHC in TNBC had many false-positives ⁹⁵ ⁹⁶. Guidelines suggest confirmatory testing by molecular methods if IHC is positive, rather than acting on IHC alone. - **ASCO** doesn't have separate guidelines but has endorsed the use of TRK inhibitors for NTRK-fusion tumors. In its 2022 Tumor-Agnostic Therapies guideline, ASCO recommended TRK inhibitor therapy for patients with NTRK fusion-positive cancers who have no satisfactory alternatives, which would apply to breast cancer as well. - **Insurance and policy:** Many payers require proof of the fusion (e.g. an NGS report) and usually that the patient has tried standard treatments. But since it's an approved indication, once criteria are met, entrectinib is generally available. - **Clinical Trials Ongoing:** While entrectinib is approved, research continues. For example, there are trials combining TRK inhibitors with other agents to prevent resistance (like combining larotrectinib with a MEK inhibitor to preempt MAPK upregulation). In breast cancer specifically, due to low incidence, no combination trial exists purely for breast, but concepts can be borrowed (e.g. a patient progressing on entrectinib with a MET amplification could be enrolled in a trial of entrectinib + crizotinib (a MET inhibitor) if available). - **Real-world data:** Some registries (like the Flatiron database or others) have started capturing outcomes of patients treated off-trial. Early real-world data suggests that responses to TRK inhibitors in NTRK-fusion breast cancers mirror trial results in efficacy, but sometimes with more dose interruptions due to toxicity management in practice (as community oncologists may be cautious with CNS side effects).

Finally, it's worth noting that **entrectinib vs larotrectinib:** There's no head-to-head trial, but a meta-analysis (Drilon et al, 2021) suggested similar efficacy. Entrectinib may have slightly more CNS adverse effects and

some lab abnormalities (transaminases, uric acid) whereas larotrectinib is very mild except for dizziness. However, entrectinib's advantage is its activity against ROS1/ALK – irrelevant for most breast cancers, but if a breast cancer patient had, say, a concurrent ALK variant of unknown significance, entrectinib covers that too. In any case, guidelines list both as options ¹⁰. Choice may depend on availability or physician experience.

In the **context of sequencing**: For metastatic breast cancer, one would typically use all standard therapies (endocrine, chemo, HER2-targeted as applicable) first. TRK inhibitors come into play either when standard options are exhausted or if the tumor is clearly driven by the fusion and is refractory to other therapy. For secretory carcinoma, some experts argue for using a TRK inhibitor early (since chemo is often not very effective in those, given their indolence). In fact, an anecdote: a pediatric secretory breast cancer case was treated with entrectinib upfront, avoiding intensive chemo, and had a great outcome ⁹⁷ ⁹⁸. So, guidelines allow flexibility – it's approved regardless of line of therapy as long as criteria are met (advanced, fusion-positive, no alternatives).

Preclinical context: The understanding of pathways in resistance (like the MEK combo, etc.) comes from preclinical studies such as a Cell Reports 2020 paper that demonstrated how combining entrectinib with cobimetinib delayed resistance in TRK fusion models ³⁷. Another preclinical note: entrectinib was originally developed by Nerviano in Italy (NMS-E628) and had potent activity in cell lines and xenografts harboring NTRK, ROS1, or ALK rearrangements, setting the stage for clinical success ⁸⁸.

Wrapping up, **entrectinib is firmly established in guidelines as the therapy of choice for any breast cancer patient with an NTRK fusion, typically to be used when the disease is metastatic or unresectable and progressing**. Its approval has expanded the precision medicine toolkit in breast oncology, albeit for a very small slice of patients. Ongoing surveillance for long-term outcomes will continue, and patients are encouraged to enroll in registries or trials (like the ASCO TAPUR study had an arm for TRK inhibitors) to contribute data. Meanwhile, next-generation inhibitors (e.g. selitrectinib, repotrectinib) are in trials (NAVIGATE, etc.) and could become options if entrectinib or larotrectinib fail due to resistance.

Given the current knowledge, if a breast cancer patient is found to have an NTRK fusion after broad genomic testing, the care team should: verify it with a high-quality assay, discuss targeted therapy, consider referring to a clinical trial (if say the patient is still on first-line therapy, one might wait; if progressed, go for TRK inhibitor), and manage the patient per guidelines (including all the safety monitoring described). The prognosis for such patients with targeted therapy is significantly improved compared to if the fusion were not identified – for example, secretory breast carcinoma metastases that would be incurable with chemo have shown complete responses on TRK inhibitors, something almost unprecedented in that context ⁷. This truly showcases the value of precision oncology in breast cancer care.

Pathway Evidence Table

Pathway (ID/Name)	Regulation in Context	Effect of Regulation on Drug Response	Biological Rationale (Breast Cancer Context)	Key References
Neurotrophin/TRK Signaling (KEGG: hsa04722)	Downregulated by entrectinib (target pathway suppressed)	Sensitive – Tumor regresses when TRK pathway is turned off	TRK fusion is the oncogenic driver; blocking TRK signaling halts proliferation and induces apoptosis in NTRK-driven breast cancer ^{15 19} . Suppression of this pathway is the primary mechanism of action, leading to tumor sensitivity.	Wu <i>et al.</i> , 2019 DrugBank M
RAS-MAPK/ERK Pathway (HALLMARK_KRAS_SIGNALING)	Upregulation (reactivation of MAPK cascade)	Resistant – Reactivated ERK signaling bypasses TRK inhibition	Upstream TRK blockade fails if MAPK pathway is turned back on via KRAS mutations or other means. E.g., acquired KRAS^G12C mutation led to ERK reactivation and entrectinib resistance ²¹ . Upregulated ERK drives proliferation independent of TRK.	Drilon <i>et al.</i> , 2019 (Cell Rep) ²¹

Pathway (ID/Name)	Regulation in Context	Effect of Regulation on Drug Response	Biological Rationale (Breast Cancer Context)	Key References
RAS-MAPK/ERK Pathway (HALLMARK_KRAS_SIGNALING)	Downregulation (sustained suppression of MAPK)	Sensitive – Tumors remain responsive when MAPK remains off	Effective TRK inhibition keeps ERK signaling low, preventing proliferation. In models, combining a MEK inhibitor kept MAPK fully suppressed and forestalled resistance , prolonging tumor regressions ³⁸ . Downstream silence of MAPK is required for durable response.	Cocco <i>et al.</i> , (Cell Rep) ³⁸
PI3K-AKT-mTOR Pathway (HALLMARK_PI3K_AKT_MTOR_SIGNALING)	Upregulation (e.g. by PIK3CA mutation or PTEN loss)	Resistant – Alternate survival signaling sustains tumor growth	Hyperactive PI3K/AKT provides a TRK-independent survival route. In one NTRK-fusion breast ca, a co-occurring <i>PIK3CA</i> E545K mutation was found ²² ; such activation of AKT can blunt apoptosis despite TRK blockade. Upregulated PI3K signaling is a known bypass resistance mechanism ¹⁶ .	Front. Mol. 2021 ²² ; DrugBank M

Pathway (ID/Name)	Regulation in Context	Effect of Regulation on Drug Response	Biological Rationale (Breast Cancer Context)	Key Referen
PI3K-AKT-mTOR Pathway (HALLMARK_PI3K_AKT_MTOR_SIGNALING)	Downregulation (no alternate PI3K activation)	Sensitive – Cells depend entirely on TRK; loss of PI3K backup leads to apoptosis	When PI3K/AKT is not independently active, TRK inhibition cuts off the main survival signal. Tumors without PIK3CA mutations or with intact PTEN (low PI3K activity) respond better to entrectinib (no parallel pathway). Experimentally, suppressing PI3K/mTOR signaling (e.g. with inhibitors) synergizes with TRK blockade	Drilon <i>et al.</i> , 2015 ²¹ ; Vaishn ⁹⁹

Pathway (ID/Name)	Regulation in Context	Effect of Regulation on Drug Response	Biological Rationale (Breast Cancer Context)	Key References
MET Signaling Pathway (REACTOME_R-HSA-8875555: MET activates RAS)	Upregulation (MET gene amplification or HGF ↑)	Resistant – MET bypasses the need for TRK, activating downstream pathways	MET amplification is a common off-target resistance: it reactivates MAPK/PI3K via MET. Off-target TRK-i resistance studies show MET amplification or overexpression in resistant tumors ²⁴ . MET up = continued growth despite TRK shutoff. Clinically, MET-amplified TRK-inhibitor-resistant cancers have been observed (requiring MET inhibitors).	Amatu <i>et al</i> (Ann Oncol)

Pathway (ID/Name)	Regulation in Context	Effect of Regulation on Drug Response	Biological Rationale (Breast Cancer Context)	Key References
Apoptosis Pathway (HALLMARK_APOPTOSIS)	Downregulation (suppressed apoptotic execution)	Resistant – Cells survive without undergoing drug-induced death	If cancer cells fail to execute apoptosis, they resist therapy. Upregulation of anti-apoptotic proteins (BCL-2, MCL1) or loss of pro-apoptotic factors (BIM) leads to persistence. E.g., TRK-fusion tumors inherently decrease apoptosis ¹⁵ ; if this state persists (or worsens via p53 loss), entrectinib cannot kill cells effectively.	Wu <i>et al.</i> , 2018 Drilon <i>et al.</i> , 2018 ⁴

Pathway (ID/Name)	Regulation in Context	Effect of Regulation on Drug Response	Biological Rationale (Breast Cancer Context)	Key References
Apoptosis Pathway (<i>HALLMARK_APOPTOSIS</i>)	Upregulation (enhanced pro-apoptotic activity)	Sensitive – Facilitates robust tumor cell death upon TRK inhibition	Entrectinib shifts balance towards apoptosis ²³ . Tumors with primed apoptotic machinery (e.g. high baseline Bax/Bim, intact TP53) undergo rapid cell death when survival signals are cut. Upregulation of cleaved caspases and apoptotic gene sets was noted in responding tumor models ²³ , correlating with deeper responses.	DrugBank Pharmacod ²³

Pathway (ID/Name)	Regulation in Context	Effect of Regulation on Drug Response	Biological Rationale (Breast Cancer Context)	Key References
EMT & Invasion Program (GO:0001837 <i>epithelial-to-mesenchymal transition</i>)	Upregulation (EMT activation, E-cadherin ↓)	Resistant – EMT state confers drug tolerance and alternative signaling	Mesenchymal transition reduces reliance on original driver and is associated with resistance to TKIs ⁴² ⁴³ . In TRK context, EMT can activate AXL, FGFR, or other pathways and decrease proliferation (drug less effective). NGF–TrkA signaling in other cancers induces EMT and drug resistance ⁴² , suggesting that if breast cancer cells undergo EMT under drug pressure, they become less killable.	Lin <i>et al.</i> , 2018 (Nature) ⁴² ; Subbiah <i>et al.</i> , 2018 ³⁹

Pathway (ID/Name)	Regulation in Context	Effect of Regulation on Drug Response	Biological Rationale (Breast Cancer Context)	Key References
NTRK Kinase Domain (Target) (NTRK3 Gene/Protein)	Downregulation (loss of fusion expression or kinase inactivation)**	Resistant – Target no longer present or bindable, so drug has nothing to act on	If the tumor stops expressing the NTRK fusion (through histologic change or promoter downregulation) or acquires a mutation that inactivates the kinase, it can survive via alternate means. E.g., a tumor switching lineage (to neuroendocrine or squamous) may silence the fusion – effectively a “target loss”. Also, on-target mutations (NTRK3 G623R etc.) prevent drug binding ³⁹ – functionally the drug can’t inhibit TRK, yielding resistance. In either case, the TRK pathway is no longer a viable target.	Russo <i>et al.</i> , ³⁹ (mechanism review); Derubeau <i>et al.</i> , 2022 ¹⁰⁷

Pathway (ID/Name)	Regulation in Context	Effect of Regulation on Drug Response	Biological Rationale (Breast Cancer Context)	Key Referer
HER2 Signaling (KEGG hsa04012: ErbB signaling)	Upregulation (HER2 amplification/activation)	Resistant – HER2 pathway drives growth despite TRK inhibition	In HER2+ breast cancers with NTRK fusions, active HER2 signaling can dominate. A breast tumor with an ERBB2–NTRK3 fusion and HER2 3+ amplification ²² would likely not respond fully to TRK inhibition alone because HER2 (ERBB2) upregulation continues activating PI3K/MAPK. The tumor remains addicted to HER2 signals. Upregulated HER2 (or EGFR) thus causes resistance unless co-inhibited.	Wang <i>et al.</i> , (Front Oncol OnLive 202

Hormone Receptor (ER) Signaling
(GO:0033144 response to steroid hormone)	Upregulation (reactivation of ER pathway)	Resistant – ER-driven proliferation provides an alternate growth mechanism	In ER+ breast cancer, if estrogen receptor signaling becomes active (through ESR1 mutations or increased estrogen etc.), the tumor can proliferate via ER target genes even if TRK is inhibited. For instance, an ER+ tumor that acquired an NTRK fusion might initially respond to TRK inhibitor, but if it then upregulates ER output (perhaps after stopping endocrine therapy), it could resume growth. Essentially, the cancer shifts back to hormone-driven mode. This upregulation of ER pathway would make it less dependent on NTRK, thus resistant to entrectinib alone.	Clinical case discussions (Vaishnavi 2012) (no direct breast case published; extrapolate)
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Table Legend: *Regulation* denotes whether the pathway's activity is increased (Up) or decreased (Down) in relation to entrectinib treatment or resistance. *Effect of drug* indicates if that pathway change is associated with a **sensitive** (good response) or **resistant** (tumor evading drug) outcome. Biological rationale provides

context for why that pathway's modulation affects entrectinib efficacy in breast cancer. Each entry is supported by literature evidence specific to human data (or closely relevant tumor models).

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